

Proposed Solution Proposal

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Proposed Solution Proposal
Project Phase	Phase 3: Project Design Phase
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Project Objective & Scope

The project automates the loan appraisal process using machine learning. Applicant information is analyzed to classify applications as approved or rejected.

2. Hardware Resource Requirements Table

Hardware Unit	Minimum Specifications Required	Project Implementation Status
CPU Processor	Multi-core processor @ 2.0 GHz or higher (x86/x64 architecture)	Compliant (Development system CPU)
RAM Memory	8 GB RAM minimum	Compliant
Storage Type	128 GB SSD with at least 15 GB available space	Compliant

3. Software & Dependency Requirements Table

Software Layer	Version Specification	Justified Implementation Need
Programming OS	Python 3.10+ (x64 version)	Runtime interpreter language
Web micro-framework	Flask v3.0+	Handles incoming HTTP requests and routes static template pages
Classifier Package	XGBoost v1.7.3 / Scikit-learn v1.2.2	Runs modeling classes and scaling objects
File Serialization	Joblib v1.2.0	Deserializes model variables immediately during application startup